BNF rules

<lolcode> ::= "#HAI' <lolcode-start> "#KBYE"

<lolcode-start> ::= <comment> <lolcode-start> | <head> <body> | <variable-def> <body> | <body-element> <body> | ε

<head> ::= "#MAEK HEAD" <head-start> "#MKAY"

<head-start> ::= <comment> <head-start> | <variable-def> <title> | <title>

<title> ::= "#GIMMEH TITLE" <words> "#OIC"

<comment>::= "#OBTW" <comment-text> "#TLDR"

<comment-text> ::= <text> <comment-text> | ε

<body> ::=<comment> <body> | <body-element> <body> | ε

<body-element> ::= <paragraph> | <bold> | <list> | <newline> | <link> | <variable-use> | <text>

<paragraph> ::="#MAEK PARAGRAF" <paragraph-start> "#MKAY"

<paragraph-start> ::= <variable-def> <inner-paragraph> | <inner-paragraph>

<inner-paragraph> ::= <comment> <inner-paragraph> | <bold> <inner-paragraph> | <list> <inner-paragraph> | <newline> <inner-paragraph> | <link> <inner-paragraph> | <text> <inner-paragraph> | <variable-use> <inner-paragraph> | ε

<bold> ::= "#GIMMEH BOLD" <inner-text> "#OIC"

<inner-text> ::= <variable-use> <inner-text> | <text> <inner-text> | ε

<list> ::= "#MAEK LIST" <list-body> "#MKAY"

<list-body> ::= <variable-def> <list-content> | <list-content>

<list-content> ::= <comment> <list-content> | <list-items>

<list-items> ::= <list-item> <list-tail>

<list-tail> ::= <list-item> <list-tail> | <comment><list-tail> | ε

<list-item> ::= "#GIMMEH ITEM" <list-item-content> "#OIC"

<list-item-content> ::= <bold> <list-item-content> | <variable-use> <list-item-content> | <text> <list-item-content> | ε

<newline> ::= "#NEWLINE"

<link> ::= '#GIMMEH LINX'<text> "#OIC"

<variable-def> ::= "#IHAZ" <text> "#ITIZ" <words> "#MKAY"

<variable-use> ::= "#GIMMEH VAR" <text> "#OIC"

<words> ::= <text> <words-tail>

<words-tail> ::= <text> <words-tail> | ε

<text> ::= TEXT

ANTLR

```
grammar sillylolspeak;


lolcode : HAI comment* (head | variableDef)? body KBYE;

head    :MAEK_HEAD comment* variableDef? title MKAY;

title     :GIMMEH_TITLE TEXT+ OIC;

comment       :OBTW TEXT* TLDR;

body    :bodyItem*;

bodyItem:comment | paragraph | bold | list | newline | link | variableUse | TEXT;

paragraph: MAEK_PARAGRAF variableDef? paragraphItem* MKAY;

paragraphItem:comment | bold | list | newline | link | variableUse | TEXT;

bold    : GIMMEH_BOLD boldItem* OIC;

boldItem: variableUse | TEXT;

list :MAEK_LIST variableDef? comment* listItem (listItem | comment)* MKAY;

listItem: GIMMEH_ITEM (bold | variableUse | TEXT)* OIC;

newline: NEWLINETAG;

link      :GIMMEH_LINX TEXT OIC;

variableDef:IHAZ TEXT ITIZ TEXT+ MKAY;

variableUse: GIMMEH_VAR TEXT OIC;

HAI     :'#HAI' | '#hai';

KBYE   :'#KBYE' | '#kbye';

OBTW :'#OBTW' | '#obtw';
```

```
TLDR   :'#TLDR' | '#tldr';

OIC    :'#OIC' | '#oic';

NEWLINETAG :'#NEWLINE' | '#newline';

IHAZ   :'#IHAZ' | '#ihaz';

ITIZ   :'#ITIZ' | '#itiz';

MKAY  :'#MKAY' | '#mkay';

MAEK_HEAD : ('#MAEK' | '#maek') (' ' | '\t' | '\n' | '\r')+ ('HEAD' |'head');

MAEK_PARAGRAF :   ('#MAEK' | '#maek') (' ' | '\t' | '\n' | '\r')+ ('PARAGRAF' |'paragraf');

MAEK_LIST :   ('#MAEK' | '#maek') (' ' | '\t' | '\n' | '\r')+ ('LIST' |'list');

GIMMEH_TITLE :      ('#GIMMEH' | '#gimmeh') (' ' | '\t' | '\n' | '\r')+ ('TITLE' |'title');

GIMMEH_BOLD :       ('#GIMMEH' | '#gimmeh') (' ' | '\t' | '\n' | '\r')+ ('BOLD' |'bold');

GIMMEH_ITEM :       ('#GIMMEH' | '#gimmeh') (' ' | '\t' | '\n' | '\r')+ ('ITEM' |'item');

GIMMEH_LINX :       ('#GIMMEH' | '#gimmeh') (' ' | '\t' | '\n' | '\r')+ ('LINX' |'linx');

GIMMEH_VAR :        ('#GIMMEH' | '#gimmeh') (' ' | '\t' | '\n' | '\r')+ ('VAR' |'var');

TEXT   :        ('a'..'z' | 'A'..'Z' | '0'..'9' | ',' | '.' | '"' | ':' | '?' | '!' | '%' | '=' | '/')+;

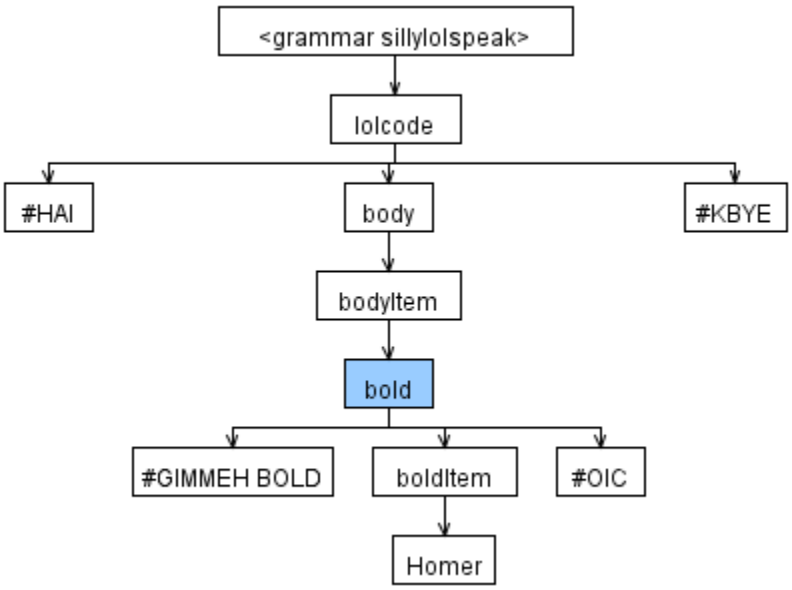
EXTRAWHITESPACE:(' ' | '\t' | '\r' | '\n')+ {skip();};
```

Testing

#HAI

#GIMMEH BOLD Homer #OIC

#KBYE



Test 2 #

#HAI
#MAEK LIST

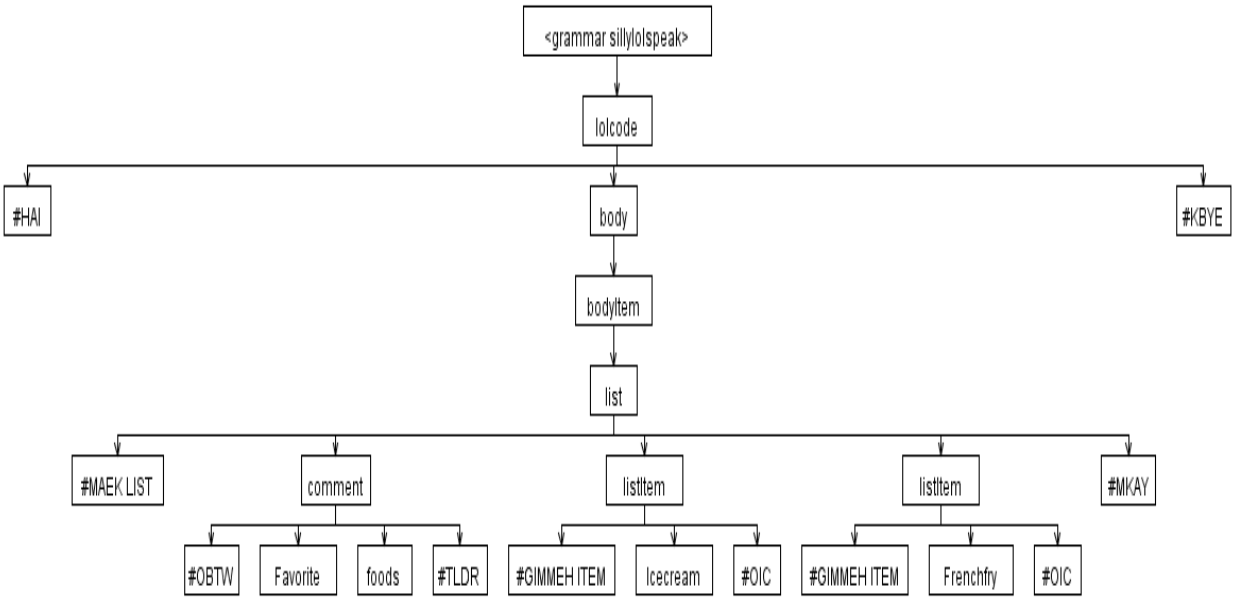
#OBTW Favorite foods #TLDR

#GIMMEH ITEM icecream #OIC

#GIMMEH ITEM Frenchfry #OIC

#MKAY

#KBYE



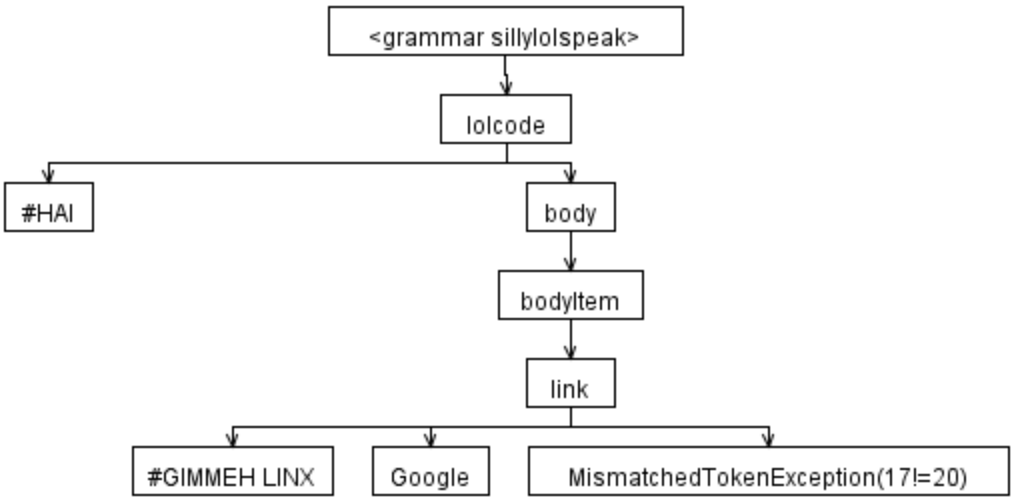
Test 3

FAILING test

#HAI

#GIMMEH LINX Google #MKAY

#KBYE


<grammar sillylolspeak>
lolcode
#HAI
body
bodyItem
link
#GIMMEH LINX
Google
MismatchedTokenException(17!=20)